

PROJECT 3: SQL DATA ANALYSIS

DecodeLabs Industrial Training | Batch 2026

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Tool: MySQL Workbench 8.0 | Database: decodelabs | Table: sales | Records: 1,200

Overview

This report presents six SQL queries run against a 1,200-row e-commerce sales dataset loaded into MySQL. Each query targets a specific business question using SELECT, WHERE, ORDER BY, GROUP BY, COUNT, SUM, and AVG. The results are shown below with the query code and output screenshot.

Query 1: View Data Sample

Purpose: Retrieve the first 10 rows to confirm the dataset loaded correctly and inspect column structure.

SQL Code:

```
USE decodelabs;
SELECT * FROM sales LIMIT 10;
```

Result:

10 rows returned. The table contains all 14 columns including OrderID, Date, Product, Quantity, UnitPrice, and TotalPrice.

MySQL Workbench interface showing the results of Query 1: View Data Sample. The query is executed against the 'decodelabs' database, and the results are displayed in the Result Grid. The table contains 10 rows of data, including columns like OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, and PaymentMethod.

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card

The bottom panel shows the action output for the query, indicating that 10 rows were returned.

Query 2: Filter Delivered Orders (WHERE)

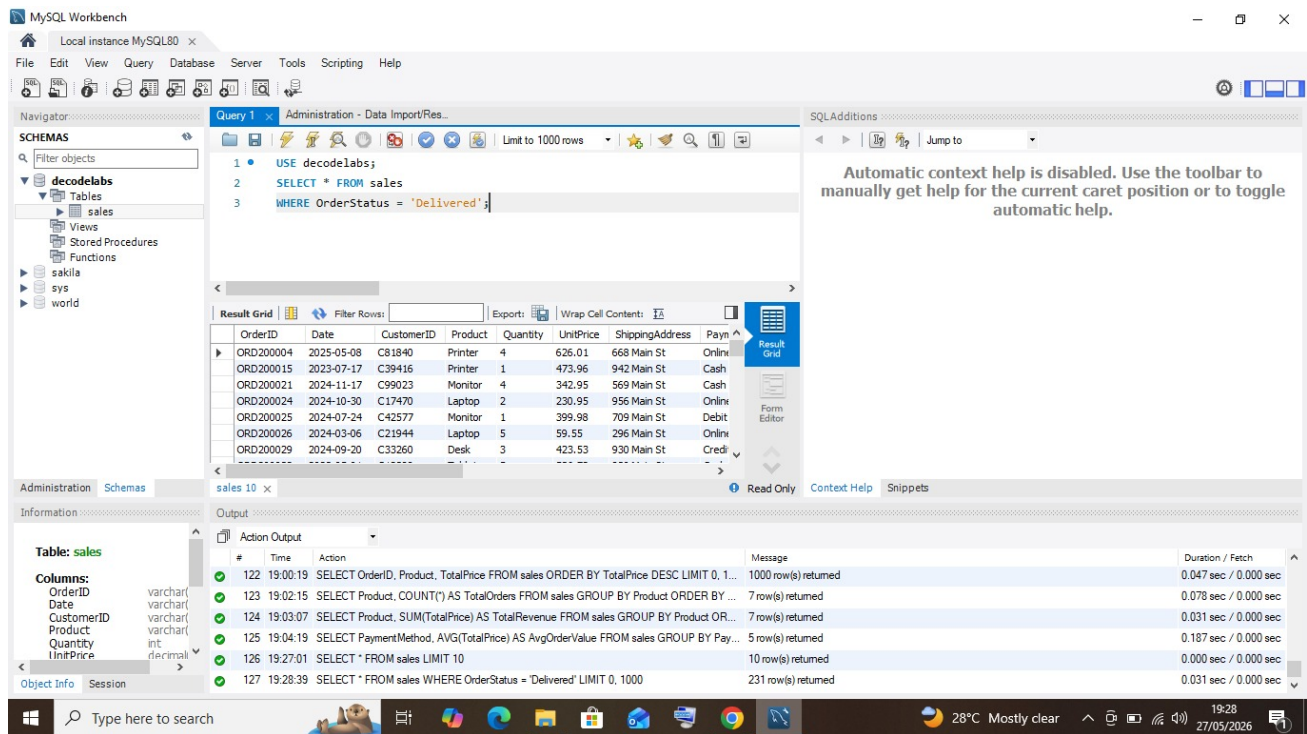
Purpose: Isolate all orders where the status is Delivered to understand fulfillment volume.

SQL Code:

```
USE decodealabs;
SELECT * FROM sales
WHERE OrderStatus = 'Delivered';
```

Result:

231 rows returned out of 1,200 total orders. This means roughly 19% of all orders were successfully delivered.



The screenshot displays the MySQL Workbench interface. The top toolbar includes options like File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar shows the 'SCHEMAS' panel with a tree view containing 'decodealabs', 'sakila', 'sys', and 'world'. The 'decodealabs' schema is expanded, showing 'Tables', 'Views', 'Stored Procedures', and 'Functions'. The 'Tables' list includes 'sales'. The main window shows 'Query 1' with the following SQL code:

```
1 USE decodealabs;
2 SELECT * FROM sales
3 WHERE OrderStatus = 'Delivered';
```

The 'Result Grid' shows the results of the query, limited to 1000 rows. The columns are: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, and PaymentMethod. The first few rows are:

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash
ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash
ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online
ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit
ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online
ORD200029	2024-09-20	C33260	Desk	3	423.53	930 Main St	Credit

The bottom panel shows the 'Action Output' with a table of query execution details:

#	Time	Action	Message	Duration / Fetch
122	19:00:19	SELECT OrderID, Product, TotalPrice FROM sales ORDER BY TotalPrice DESC LIMIT 0, 1...	1000 row(s) returned	0.047 sec / 0.000 sec
123	19:02:15	SELECT Product, COUNT(*) AS TotalOrders FROM sales GROUP BY Product ORDER BY ...	7 row(s) returned	0.078 sec / 0.000 sec
124	19:03:07	SELECT Product, SUM(TotalPrice) AS TotalRevenue FROM sales GROUP BY Product OR...	7 row(s) returned	0.031 sec / 0.000 sec
125	19:04:19	SELECT PaymentMethod, AVG(TotalPrice) AS AvgOrderValue FROM sales GROUP BY Pay...	5 row(s) returned	0.187 sec / 0.000 sec
126	19:27:01	SELECT * FROM sales LIMIT 10	10 row(s) returned	0.000 sec / 0.000 sec
127	19:28:39	SELECT * FROM sales WHERE OrderStatus = 'Delivered' LIMIT 0, 1000	231 row(s) returned	0.031 sec / 0.000 sec

Query 3: Sort Orders by Total Price (ORDER BY)

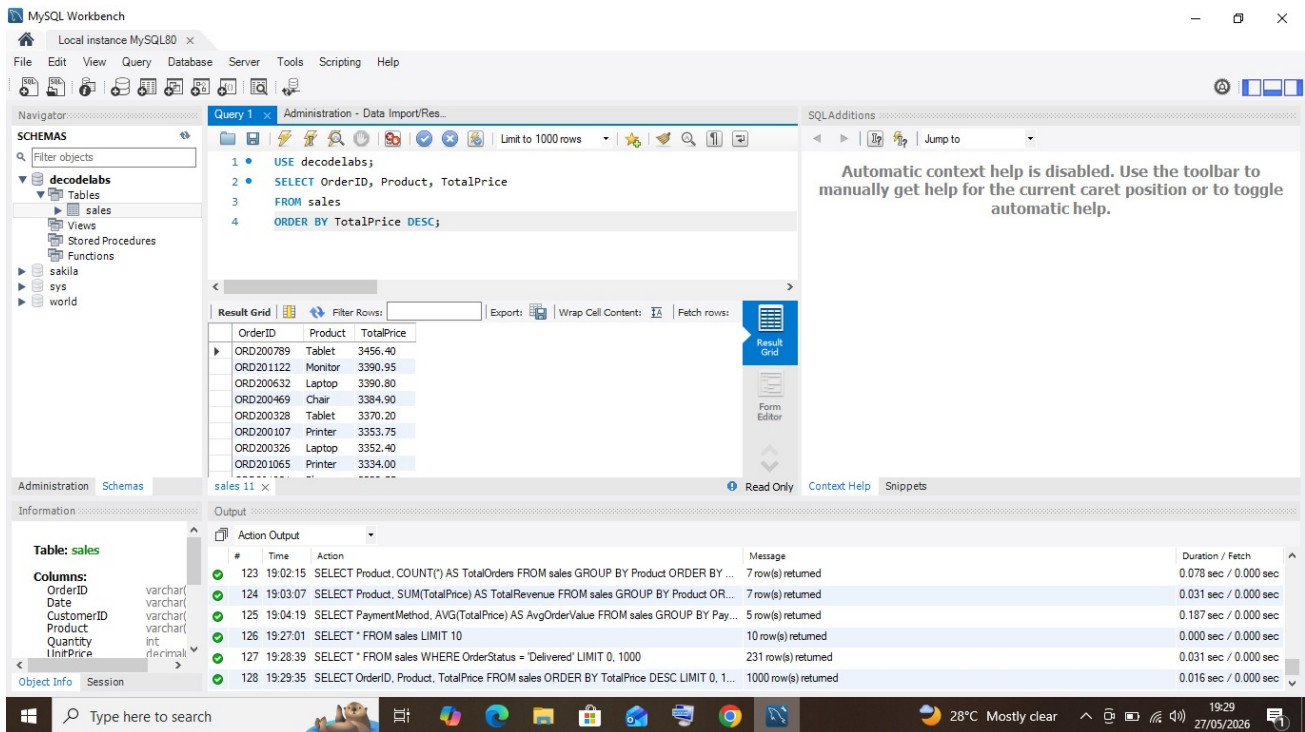
Purpose: Rank all orders from highest to lowest value to identify the most valuable transactions.

SQL Code:

```
USE decodealabs;
SELECT OrderID, Product, TotalPrice
FROM sales
ORDER BY TotalPrice DESC;
```

Result:

1,000 rows returned. The highest-value order was a Tablet (ORD200789) at 3,456.40. Tablets and Monitors dominate the top of the list.



Query 4: Count Orders per Product (COUNT + GROUP BY)

Purpose: Count how many orders exist for each product type to identify the most ordered product.

SQL Code:

```
USE decodealabs;
SELECT Product, COUNT(*) AS TotalOrders
FROM sales
GROUP BY Product
ORDER BY TotalOrders DESC;
```

Result:

7 product categories found. Printer leads with 181 orders, followed by Tablet (179) and Chair (178). Phone has the fewest at 156.

The screenshot shows the MySQL Workbench interface. The 'Query Editor' window contains the following SQL query:

```

1 • USE decodeLabs;
2 • SELECT Product, COUNT(*) AS TotalOrders
3 FROM sales
4 GROUP BY Product
5 ORDER BY TotalOrders DESC;

```

The 'Result Grid' window displays the following data:

Product	TotalOrders
Printer	181
Tablet	179
Chair	178
Laptop	173
Desk	170
Monitor	163
Phone	156

The 'Output' window shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
124	19:03:07	SELECT Product, SUM(TotalPrice) AS TotalRevenue FROM sales GROUP BY Product OR...	7 row(s) returned	0.031 sec / 0.000 sec
125	19:04:19	SELECT PaymentMethod, AVG(TotalPrice) AS AvgOrderValue FROM sales GROUP BY Pay...	5 row(s) returned	0.187 sec / 0.000 sec
126	19:27:01	SELECT * FROM sales LIMIT 10	10 row(s) returned	0.000 sec / 0.000 sec
127	19:28:39	SELECT * FROM sales WHERE OrderStatus = 'Delivered' LIMIT 0, 1000	231 row(s) returned	0.031 sec / 0.000 sec
128	19:29:35	SELECT OrderID, Product, TotalPrice FROM sales ORDER BY TotalPrice DESC LIMIT 0, 1...	1000 row(s) returned	0.016 sec / 0.000 sec
129	19:30:56	SELECT Product, COUNT(*) AS TotalOrders FROM sales GROUP BY Product ORDER BY ...	7 row(s) returned	0.031 sec / 0.000 sec

Query 5: Total Revenue per Product (SUM + GROUP BY)

Purpose: Calculate total revenue generated by each product to identify the highest earning category.

SQL Code:

```

USE decodeLabs;
SELECT Product, SUM(TotalPrice) AS TotalRevenue
FROM sales
GROUP BY Product
ORDER BY TotalRevenue DESC;

```

Result:

Chair generated the most revenue at 195,620.11, closely followed by Printer at 195,612.61. Phone had the lowest revenue at 151,722.39.

The screenshot shows the MySQL Workbench interface. The 'Query Editor' contains the following SQL code:

```

1 • USE decodealabs;
2 • SELECT Product, SUM(TotalPrice) AS TotalRevenue
3 FROM sales
4 GROUP BY Product
5 ORDER BY TotalRevenue DESC;

```

The 'Result Grid' displays the following data:

Product	TotalRevenue
Chair	195620.11
Printer	195612.61
Laptop	192126.56
Tablet	186568.95
Monitor	175651.41
Desk	167459.93
Phone	151722.39

The 'Output' pane shows the execution log with the following messages:

#	Time	Action	Message	Duration / Fetch
125	19:04:19	SELECT PaymentMethod, AVG(TotalPrice) AS AvgOrderValue FROM sales GROUP BY Pay...	5 row(s) returned	0.187 sec / 0.000 sec
126	19:27:01	SELECT * FROM sales LIMIT 10	10 row(s) returned	0.000 sec / 0.000 sec
127	19:28:39	SELECT * FROM sales WHERE OrderStatus = 'Delivered' LIMIT 0, 1000	231 row(s) returned	0.031 sec / 0.000 sec
128	19:29:35	SELECT OrderID, Product, TotalPrice FROM sales ORDER BY TotalPrice DESC LIMIT 0, 1...	1000 row(s) returned	0.016 sec / 0.000 sec
129	19:30:56	SELECT Product, COUNT(*) AS TotalOrders FROM sales GROUP BY Product ORDER BY ...	7 row(s) returned	0.031 sec / 0.000 sec
130	19:32:21	SELECT Product, SUM(TotalPrice) AS TotalRevenue FROM sales GROUP BY Product OR...	7 row(s) returned	0.016 sec / 0.000 sec

Query 6: Average Order Value by Payment Method (AVG + GROUP BY)

Purpose: Calculate the average order value for each payment method to understand customer spending by payment type.

SQL Code:

```

USE decodealabs;
SELECT PaymentMethod, AVG(TotalPrice) AS AvgOrderValue
FROM sales
GROUP BY PaymentMethod
ORDER BY AvgOrderValue DESC;

```

Result:

5 payment methods found. Credit Card customers spend the most on average at 1,127.55 per order. Debit Card customers have the lowest average at 1,001.56.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

decodeLabs

Tables

sales

Views

Stored Procedures

Functions

sakila

sys

world

Query 1 Administration - Data Import/Res...

1 USE decodeLabs;

2 SELECT PaymentMethod, AVG(TotalPrice) AS AvgOrderValue

3 FROM sales

4 GROUP BY PaymentMethod

5 ORDER BY AvgOrderValue DESC;

Result Grid

PaymentMethod	AvgOrderValue
Credit Card	1127.553974
Gift Card	1070.973565
Cash	1056.041829
Online	1017.220698
Debit Card	1001.556810

Information

Table: sales

Columns:

OrderID varchar(10)

Date varchar(10)

CustomerID varchar(10)

Product varchar(50)

Quantity int(4)

UnitPrice decimal(10,2)

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
120	18:57:06	SELECT * FROM sales LIMIT 10	10 row(s) returned	0.000 sec / 0.000 sec
121	18:58:54	SELECT * FROM sales WHERE OrderStatus = 'Delivered' LIMIT 0, 1000	231 row(s) returned	0.031 sec / 0.016 sec
122	19:00:19	SELECT OrderID, Product, TotalPrice FROM sales ORDER BY TotalPrice DESC LIMIT 0, 1...	1000 row(s) returned	0.047 sec / 0.000 sec
123	19:02:15	SELECT Product, COUNT(*) AS TotalOrders FROM sales GROUP BY Product ORDER BY ...	7 row(s) returned	0.078 sec / 0.000 sec
124	19:03:07	SELECT Product, SUM(TotalPrice) AS TotalRevenue FROM sales GROUP BY Product OR...	7 row(s) returned	0.031 sec / 0.000 sec
125	19:04:19	SELECT PaymentMethod, AVG(TotalPrice) AS AvgOrderValue FROM sales GROUP BY Pay...	5 row(s) returned	0.187 sec / 0.000 sec

28°C Mostly clear 19:26 27/05/2026

Summary of Findings

The following business insights emerged from the six SQL queries:

1. The dataset contains 1,200 orders across 7 product categories and 5 payment methods.
2. Only 231 orders (19%) have a Delivered status, which may indicate a high rate of returns, cancellations, or pending orders worth investigating.
3. Tablets appear among the highest-value individual orders, reaching up to 3,456.40 per transaction.
4. Printers are ordered most frequently (181 orders), but Chairs generate the most total revenue (195,620.11). This gap suggests Chairs carry a higher unit price despite fewer orders.
5. Credit Card is the highest-value payment method by average order, while Debit Card customers spend the least on average. This pattern could inform payment-specific promotions.
6. Phone is both the least ordered product and the lowest revenue generator, making it a candidate for marketing review.